

Ali Sufiyan Khan

📍 VIT Vellore | 📞 +91 91755 69485 | ✉ alizaydsab@gmail.com

🌐 LinkedIn | 🐙 GitHub | 📁 Portfolio

Education

Vellore Institute of Technology (VIT Vellore), B.Tech in Computer Science and Engineering Aug 2023 – May 2027

- **Relevant Coursework:** Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Artificial Intelligence

Projects

Deep Reinforcement Learning for 6DOF Robotic Arm Pick-and-Place Control

Python | PyTorch | NVIDIA Isaac Lab | Isaac Sim | PPO

- Designed a **UR10e 6DOF** pick-and-place system with a Robotiq 2F-85 gripper in NVIDIA Isaac Lab and Isaac Sim.
- Implemented PPO training across staged reaching, grasping, lifting, and placement tasks; completed **Phase 0 reach training**.
- Built vectorized simulation environments with custom observations, reward shaping, collision handling, and curriculum progression.

Hybrid Network Intrusion Detection & Anomaly Analysis Platform

Python | FastAPI | Scikit-learn | Redis | PostgreSQL | React | Docker

- Engineered a hybrid NIDS combining calibrated classification, Isolation Forest, and autoencoder-based anomaly detection with real-time incident correlation.
- Optimized batched detector throughput from **153.5 to 3,496.8 flows/s** on a **2,000-flow benchmark**; the durable Redis-to-PostgreSQL pipeline completed at **78.8 flows/s with zero final queue lag**.
- Built a containerized FastAPI/React platform with authenticated WebSockets, deterministic PCAP replay, and validation across **238 tests with 84% backend coverage**.

Hybrid Multi-Cloud Kubernetes Orchestration Platform

K3s | Docker | AWS | GCP | Tailscale | Cloudflare Tunnels | Locust

- Designed a hybrid K3s cluster integrating on-premise edge, AWS, and GCP nodes over Tailscale networking.
- Implemented cloud bursting and secured public service exposure through Cloudflare Tunnels.
- Sustained **1,000+ concurrent users** with a **0% failure rate** in distributed Locust load testing.

Distributed Job Discovery & Ranking Platform

Next.js | TypeScript | Rust | NATS JetStream | PostgreSQL | Docker

- Engineered a distributed job-discovery platform and canonical identity pipeline, retaining and normalizing **3,128 existing job records with no data loss**.
- Built persisted full-index search sessions with ranking, provenance, coverage, and feedback; verified **250 ranked jobs** while reporting **20/32 successful-source coverage**.
- Reduced the initial production response from approximately **24 seconds to 57 milliseconds** by serving indexed results before slow supplemental feeds.

Skills

Languages: Python, Rust, TypeScript, JavaScript, C, Java, SQL

AI/ML & Frameworks: PyTorch, TensorFlow, Scikit-learn, NVIDIA Isaac Lab, NVIDIA Isaac Sim, Stable Baselines3, Gymnasium, OpenCV

Systems, Web & Cloud: FastAPI, React, PostgreSQL, Redis, NATS JetStream, Kubernetes, Docker, AWS (EC2), GCP, Linux, Tailscale, Cloudflare Tunnels

Developer Tools: Git, GitHub, OpenAI Codex, Antigravity, Claude Code, Hermes Agent, JetBrains IDEs

Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

Aug 2025

- Issued by Oracle University | Valid through August 2027